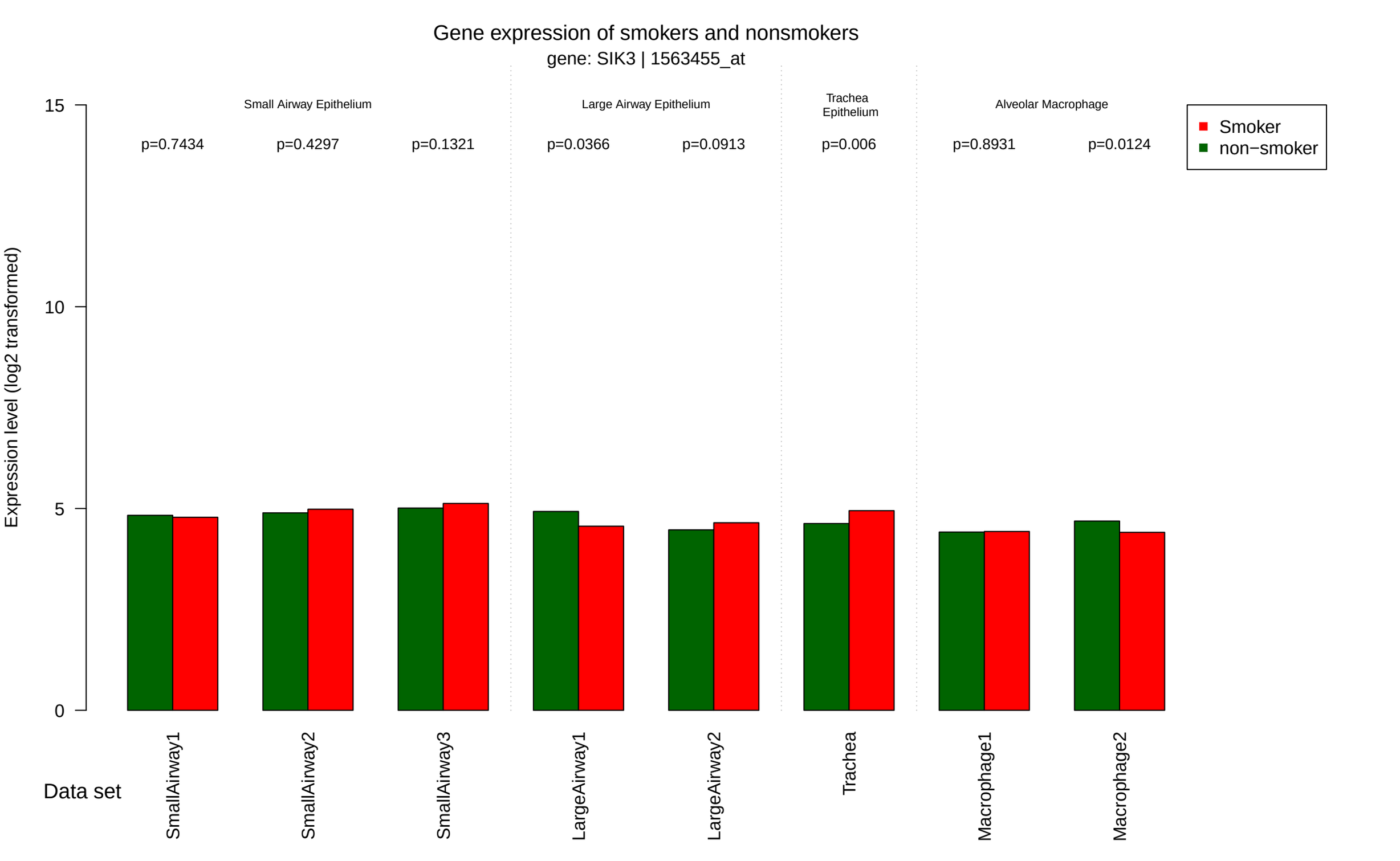
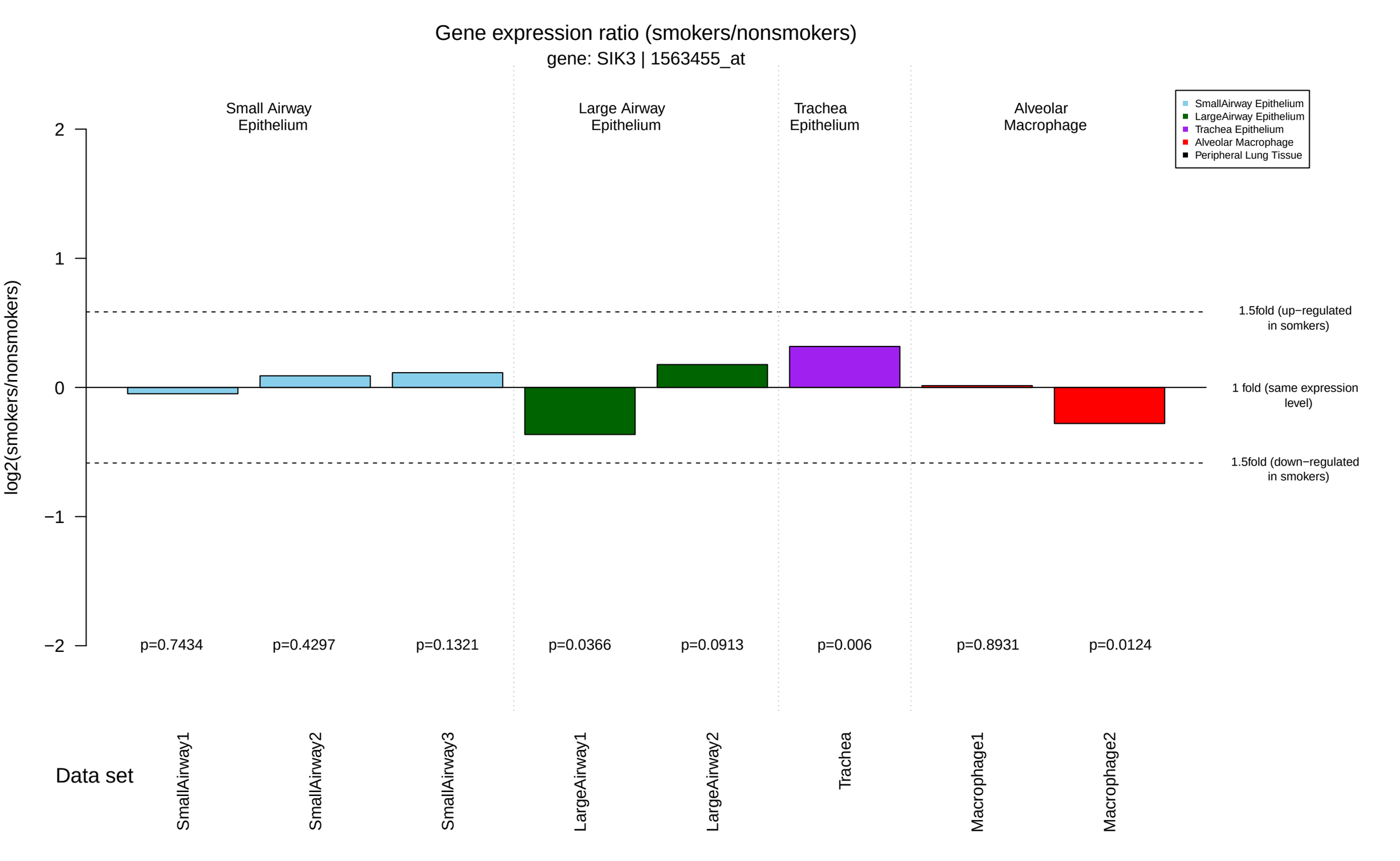
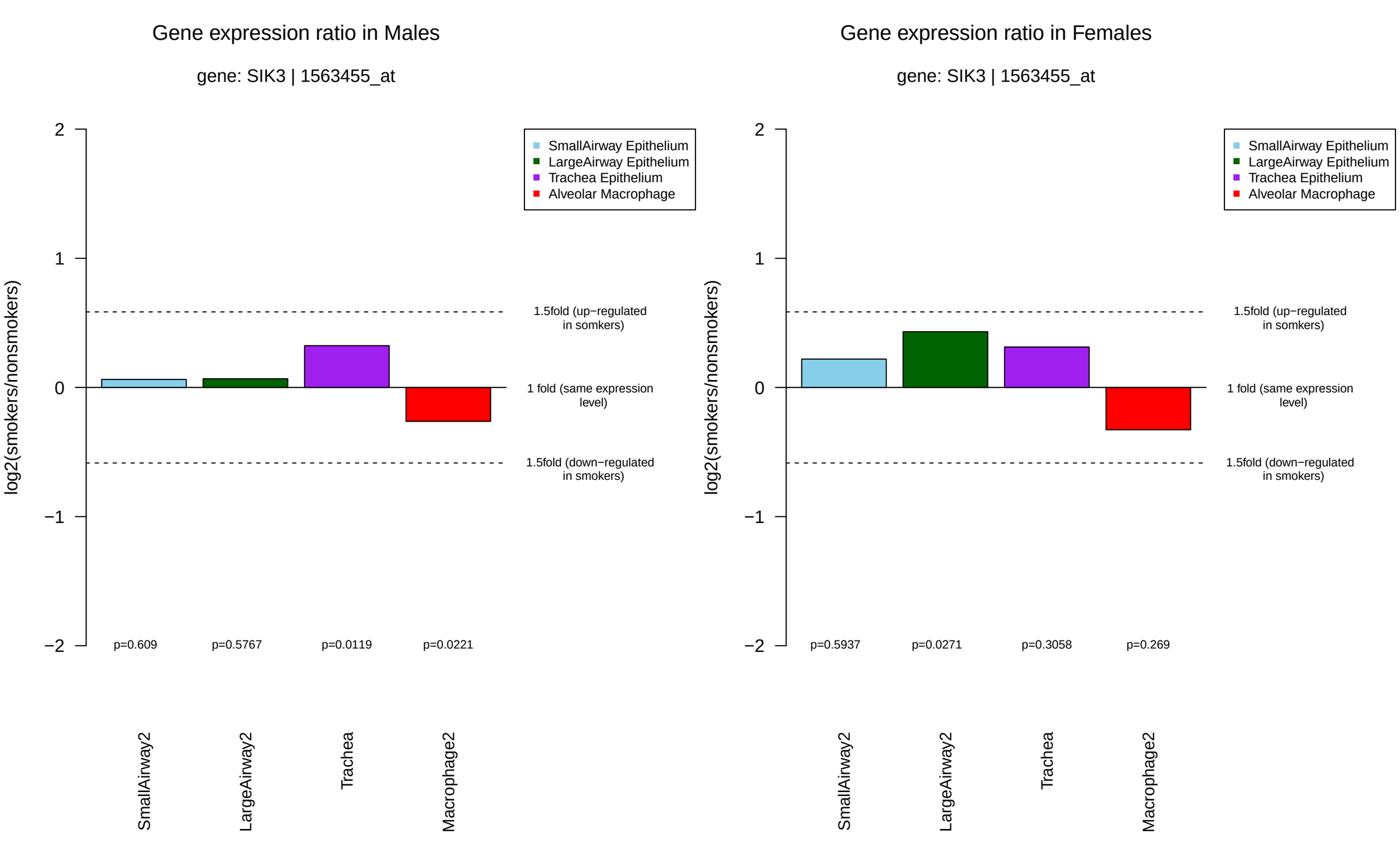
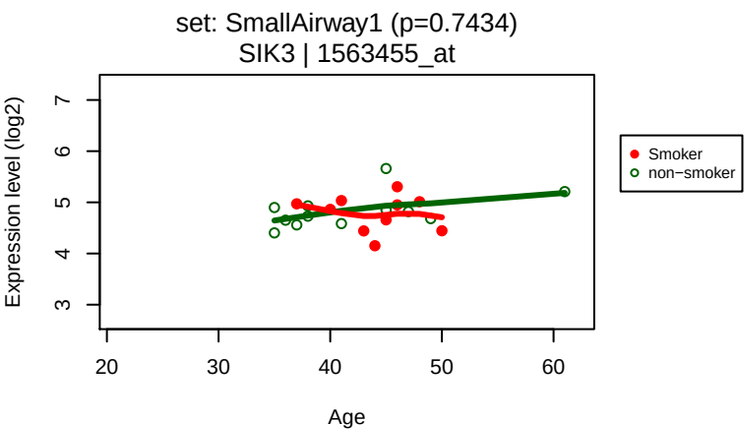
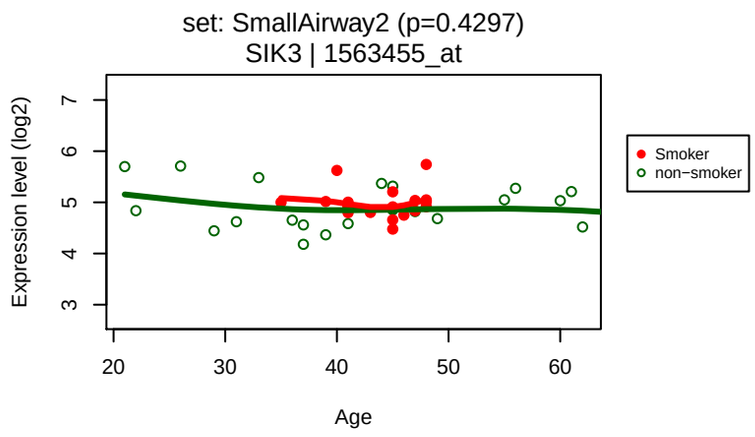
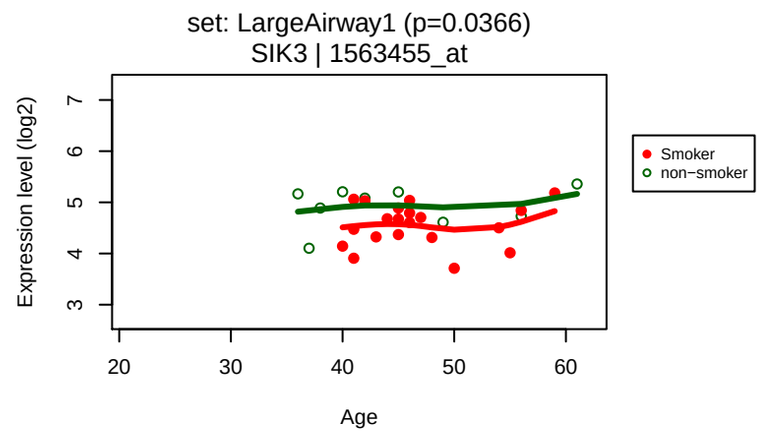

Gene expression of smokers and nonsmokers
gene: SIK3 | 1563455_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7434
p=0.4297
p=0.1321
p=0.0366
p=0.0913
p=0.006
p=0.8931
p=0.0124
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

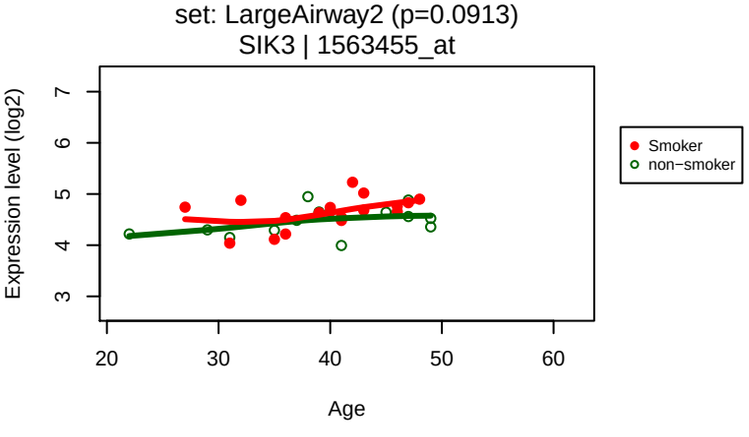

Gene expression ratio (smokers/nonsmokers)
gene: SIK3 | 1563455_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7434
p=0.4297
p=0.1321
p=0.0366
p=0.0913
p=0.006
p=0.8931
p=0.0124
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

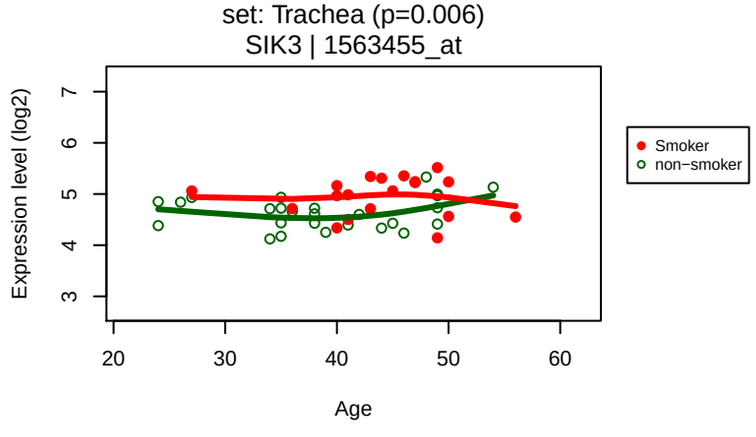

Gene expression ratio in Males
gene: SIK3 | 1563455_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.609
p=0.5767
p=0.0119
p=0.0221
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SIK3 | 1563455_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5937
p=0.0271
p=0.3058
p=0.269
SmallAirway2
LargeAirway2
Trachea
Macrophage2

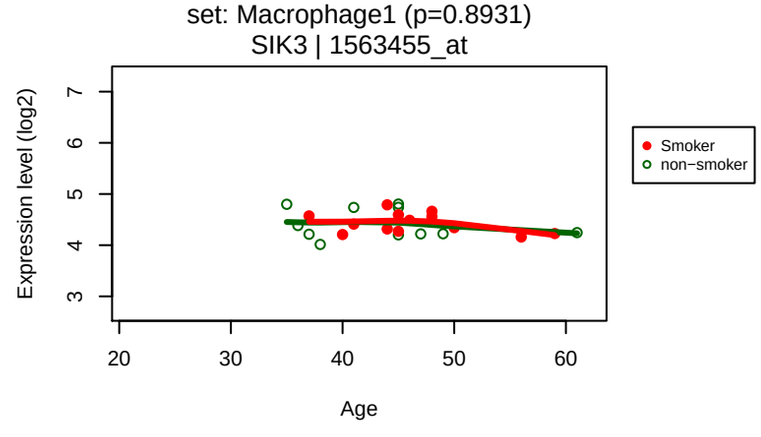

set: SmallAirway1 (p=0.7434)
SIK3 | 1563455_at
Expression level (log2)
Age
Smoker
non−smoker

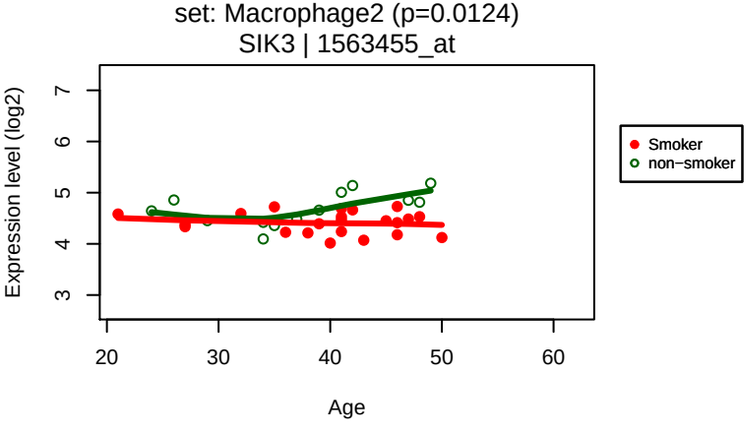

set: SmallAirway2 (p=0.4297)
SIK3 | 1563455_at
Expression level (log2)
Age
Smoker
non−smoker

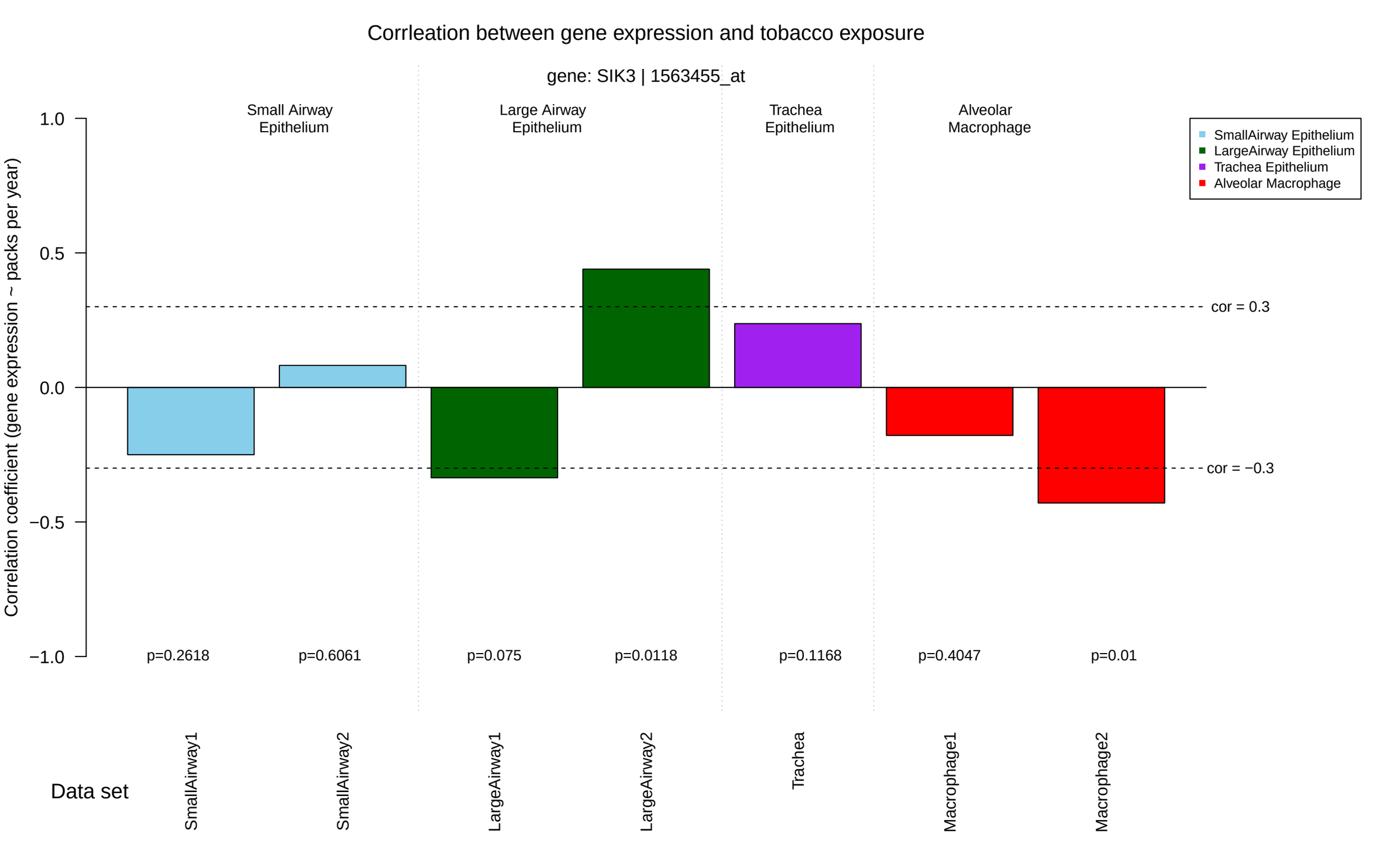

set: LargeAirway1 (p=0.0366)
SIK3 | 1563455_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0913)
SIK3 | 1563455_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.006)
SIK3 | 1563455_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8931)
SIK3 | 1563455_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0124)
SIK3 | 1563455_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SIK3 | 1563455_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2618
p=0.6061
p=0.075
p=0.0118
p=0.1168
p=0.4047
p=0.01
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2